Preliminary Research Findings:
Optimizing Byzantine Fault Tolerance in EV-to-Grid Energy Trading

Atharv Manojkumar Shukla

June 1, 2026

1 Research Objective

This research investigates the security vulnerabilities inherent in bidirectional energy exchange between Electric Vehicles (EVs) and the Distribution Network (DN). The study focuses on mitigating cyber-threats through a decentralized peer-to-peer (P2P) blockchain framework utilizing Byzantine Fault Tolerance (BFT) consensus mechanisms[cite: 5, 13].

2 Summary of Base Literature

The baseline research models the EV-DN trading ecosystem as a Byzantine General Problem (BGP), where the DN functions as the 'commander' and EVs as 'lieutenants'. The proposed security framework:

- Implements a blockchain-based immutable ledger for transaction records[cite: 100, 107].
- Employs the Oral Message Algorithm, $OM(m)$, to verify block authenticity[cite: 101, 126].
- Validates security improvements via IEEE 33-bus test feeder simulations, establishing that a successful attack requires the manipulation of $> 33\%$ of network information[cite: 18, 103].

3 Critical Analysis and Limitations

While the baseline provides a robust theoretical security foundation, the use of the $OM(m)$ algorithm presents significant practical limitations for production-grade deployments:

1. **Exponential Time Complexity:** The $OM(m)$ algorithm exhibits a communication complexity of $O(n^m)$. This recursive message-passing structure leads to severe latency as the number of nodes (n) increases[cite: 130].
2. **Scaling Bottleneck:** In high-throughput environments (e.g., thousands of simultaneous EV transactions), the recursive broadcast requirement will induce prohibitive network overhead.
3. **Simulation vs. Reality:** The current validation (50 EVs) is insufficient to represent the transaction density of a modern smart city grid.

4 Proposed Modernized Consensus Architectures

To address these scaling bottlenecks, we propose transitioning from legacy $OM(m)$ to high-performance consensus mechanisms.

Table 1: Comparative Analysis of Consensus Mechanisms

Mechanism	**Complexity**	**Deployment Viability**
Classical PBFT	$O(n^2)$	High latency; leader bottleneck.
DAG-Based (Gossip)	Variable	Bandwidth-intensive.
Sharding-BFT	Linear/Cluster	Optimal for parallel localized grids.
Tower BFT	$O(n)$	**Recommended;** utilizes Proof-of-History.

5 Research Direction: Recommended Upgrade

We suggest implementing **Tower BFT**, an optimized consensus architecture that leverages Proof-of-History (PoH) as an embedded cryptographic clock. This architecture:

- Decouples chronological event-ordering from the consensus voting process.
- Reduces message overhead from exponential/quadratic scaling to linear ($O(n)$).
- Provides deterministic finality, enabling high-frequency energy state transitions essential for dynamic grid load balancing.